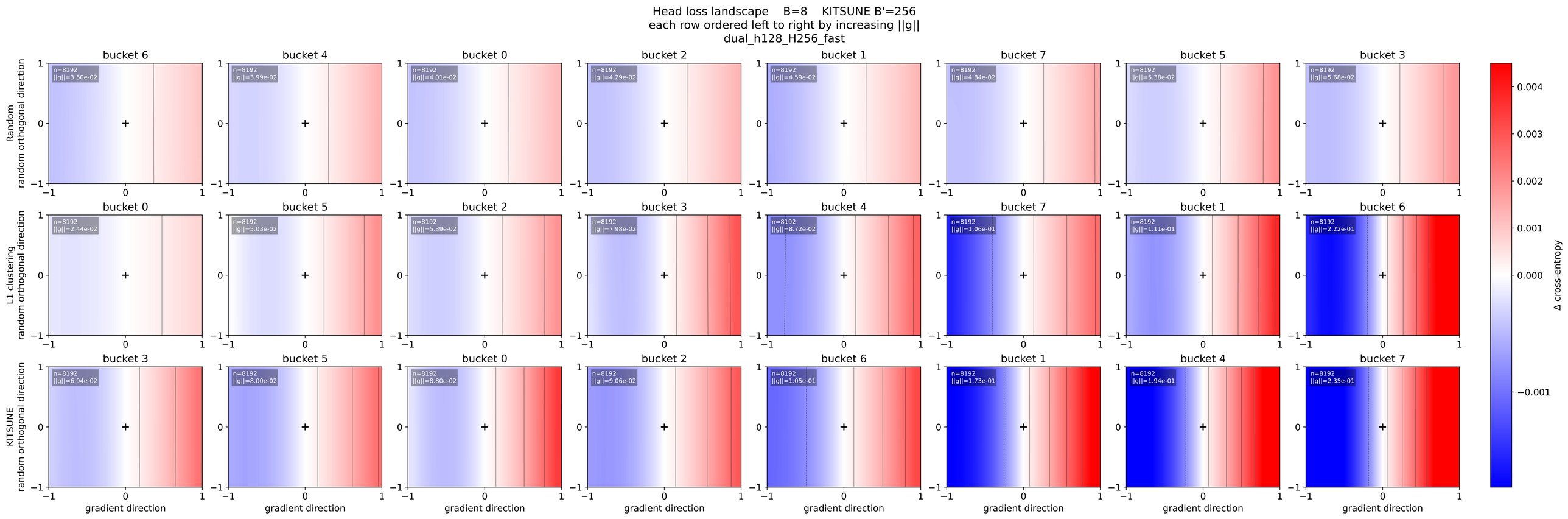

Head loss landscape B=8 KITSUNE B'=256
each row ordered left to right by increasing ||g||
dual_h128_H256_fast
Random
random orthogonal direction
L1 clustering
random orthogonal direction
KITSUNE
random orthogonal direction
bucket 6
n=8192
||g||=3.50e-02
bucket 4
n=8192
||g||=3.99e-02
bucket 0
n=8192
||g||=4.01e-02
bucket 2
n=8192
||g||=4.29e-02
bucket 1
n=8192
||g||=4.59e-02
bucket 7
n=8192
||g||=4.84e-02
bucket 5
n=8192
||g||=5.38e-02
bucket 3
n=8192
||g||=5.68e-02
bucket 0
n=8192
||g||=2.44e-02
bucket 5
n=8192
||g||=5.03e-02
bucket 2
n=8192
||g||=5.39e-02
bucket 3
n=8192
||g||=7.98e-02
bucket 4
n=8192
||g||=8.72e-02
bucket 7
n=8192
||g||=1.06e-01
bucket 1
n=8192
||g||=1.11e-01
bucket 6
n=8192
||g||=2.22e-01
bucket 3
n=8192
||g||=6.94e-02
bucket 5
n=8192
||g||=8.00e-02
bucket 0
n=8192
||g||=8.80e-02
bucket 2
n=8192
||g||=9.06e-02
bucket 6
n=8192
||g||=1.05e-01
bucket 1
n=8192
||g||=1.73e-01
bucket 4
n=8192
||g||=1.94e-01
bucket 7
n=8192
||g||=2.35e-01
gradient direction
Δ cross-entropy
0.004
0.003
0.002
0.001
0.000
−0.001